1 ECE 221A Notes

1.1 Energy band of particle in free space

Band Gap is the Minimum difference between the Conduction band and the Valence band.
E-K curve is like a parabola, the larger the lattice constant, the steeper.
Electron has mass 9.109×10^{-31} kg and kt/q ≈ 25.85 mV, $\hbar = h/\pi = 1.055 \times 10^{-34} Js$ or $6.626/\pi \times 10^{-34} Js$

$$E = \frac{\hbar^2 k^2}{2m} \qquad m^* = \hbar / \frac{d^2 E}{dk^2}$$

de Broglie relation, momentum p, and angular vector k, velocity v (equations have negative sign for holes)

$$\vec{p} = \hbar \vec{k} \qquad \vec{v} = -\frac{1}{\hbar}\frac{dE}{dk} = \frac{\vec{p}}{m^*}$$

1.2 Fermi-Level

Fermi-Dira Distribution

$$f = \left[1 + e^{(E-E_f)/kT}\right]^{-1}$$

Number of Electrons in Conduction Band (m^* is the effective mass) Left is Boltzmann approximation. N_c/N_v means the effective density of states in conduction/valence band.

$$n = N_c exp\left(\frac{E_F - E_C}{kT}\right) \quad N_c = \frac{m_e^{*3/2}}{\sqrt{2\pi^3}\hbar^{3/2}}(kt)^{1.5}$$
$$p = N_v exp\left(\frac{E_v - E_F}{kT}\right) \quad N_v = \frac{m_h^{*3/2}}{\sqrt{2\pi^3}\hbar^{3/2}}(kt)^{1.5}$$

np relation, If intrinsic $n = p = \sqrt{np}$, (means there's the same amount of electrons and holes)

$$np = n_i^2 = N_c N_v exp\left(\frac{-E_c + E_v}{kT}\right)$$

We can dope to form n/p types, ionized particles have densities, N_D/N_A doped donor/acceptor density.

$$N_D^+ = \frac{N_D}{1 + 2e^{(E_F - E_D)/kT}} \quad N_A^+ = \frac{N_A}{1 + 4e^{(E_A - E_F)/kT}}$$

Charge Neutrality is always valid

$$n + N_A^+ = p + N_D^+ \qquad np = n_i^2$$

Shallow Dopant Approximation

$$n = N_D \qquad p = N_A$$

1.3 Current Continuity

Drift Carrier Velocity

$$\vec{F} = m_e^* \vec{a} = e\vec{E} \qquad v_{avg} = \frac{e\tau}{m_e^*}\vec{E} = \mu\vec{E}$$

Drift Current and Current Density

$$I = \frac{dQ}{dt} = \frac{qnAdx}{dt} = Aqnv = A\sigma_e E$$
$$J = \frac{I}{A} = qn_e\mu_e E = \sigma E$$

Diffusion Current (l electron mean free path, n(z) electron concentration at z)

$$J = \Delta J = qlv_{th}\frac{dn}{dx}$$

Total Current Density = Drift + Diffusion

$$J_n = J_{n,drift} + J_{n,drift} = q\mu_n n\vec{E} + qD_n\frac{dn}{dx}$$
$$J_p = J_{p,drift} + J_{p,drift} = q\mu_p p\vec{E} - qD_p\frac{dp}{dx}$$

1.4 Generation and Recombination

Recombination

1. **Band-To-Band**, electrons falls back from conduction into valence
2. **Trap-Assisted**, Trapped Electrons falls back to valence
3. **Augur**, Recombination Band-To-Band energy given off to third particle

Generation

1. **Photon Absorption**photon energy enough to lift e- out of valence
2. **Energy Beam** High energy beam ionize the semiconductor
3. **Impact Ionization** e-(h+) with energy much larger(smaller) than the conduction(valence) band edge given off to generate an e-h pair through a band to band transition

1.5 Traps

Probability of Trap Occupancy follows the Fermi-Dirac distribution. Since N_t measures the trap density. For capture $N_t f(E_t)$ and emit $N_t(1 - f(E_t))$ measures the average number of filled/empty defect sites that can propagate to.

$$f(E_t) = \left[1 + e^{(E_t - E_f)/kT}\right]^{-1}$$

Electron and hole Capture and emission rate (e- density n, trap density N_t, $\sigma(cm^{-2})$ cross section, e_n and e_p emission prob unknown), v_{th} is thermal moving Net combination will be $r_{capt} - r_{emit}$

$$r_{e,capt} = v_{th}\sigma n N_t(1 - f) \quad r_{e,emit} = e_n N_t f$$
$$r_{h,emit} = e_p N_t(1 - f) \quad r_{h,capt} = v_{th}\sigma p N_t f$$

Emission Rate = Capture Rate at equilibrium, so

$$e_n = v_{th}\sigma_n n_i exp\left(\frac{E_t - E_i}{kT}\right)$$
$$e_p = v_{th}\sigma_p n_i exp\left(\frac{E_i - E_t}{kT}\right)$$

Exchanges with the Conduction/Valence band are called the emission or capture of electrons/holes.

1.6 PN Junction

Formed by P-Type N-Type Contact

1. h+ diffuse p to n, e- diffuse n to p
2. Carriers depleted at the boundary forming Depletion Region
3. E-Field formed in Depletion Region, causes the Carriers to Drift
4. Drift and Diffusion Balances
5. Slope is proportional to the doping density, Direction of E-Field determined by the sign of the gradient

Built-In Voltage $V_{bi}(eV)$ depends only on doping concentration and the bandgap.

$$V_{bi} = \frac{kT}{q}\ln\left(\frac{N_d N_a}{n_i^2}\right) = \int_{-\infty}^{\infty} E(x)dx$$
$$V_{ni} = -\frac{1}{2}(W_p + W_n)\frac{qN_D}{\varepsilon_r\varepsilon_0}W_n$$

We can also understand this as the change it has to made to align the two vacuum levels. In the case of MOSCAP this is called surface potential. If there's no overall difference across the two end that means there will not be any built-in voltage

1.6.1 Depletion Width

If the contact is at 0, We can said that the depletion into N is a negative number x_N and the depletion into p x_P. Each will have its own constant charge density ρ. $E = \int \rho dx$, $V = \int E dx$ Approximately

$$V(x) = \begin{cases} -\frac{qN_d}{2\varepsilon_s}(x - x_N)^2 + V_{bi} & x_N \le x \le 0 \\ -\frac{qN_a}{2\varepsilon_s}(x_P - x)^2 & 0 \le x \le x_P \end{cases}$$

The width can be calculated from V_{bi}

$$W_n N_D = W_p N_A \qquad W = \sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_A + N_D}{N_A N_D}}$$

Approximations

$$N_a \gg N_d \qquad W_{dep} \approx |x_N| \approx \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_d}}$$
$$N_a \ll N_d \qquad W_{dep} \approx |x_P| \approx \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_a}}$$

Under Reverse Bias $V = -|V_r|$ (Reverse bias adds to the potential barrier, negative voltage would bring down fermi level because you now loss the electrons with a higher energy level so they can only be found at lower levels)

$$W_{dep} \approx \sqrt{\frac{2\varepsilon_s(V_{bi} - V)}{(qN_{a/d})}}$$

If $N_A \gg N_D$, $W \approx W_n$, reverse bias ($V = -|V_r|$) and forward bias ($V = V_f$). The width also produces Capacitance, in units F/cm^2

$$C_J = \frac{\varepsilon_0\varepsilon_r}{W} \qquad C_{diff} = \frac{2q}{3kT}I_{dc}\tau_p$$

Assume low injection, so use quasi-fermi level concept, Boltzmann approximation still valid, external potential drops mainly across depletion region because the pn dipole is the major barrier. There's no recombination in depletion region.

1.6.2 Quasi fermi level

1. Electrons and Holes across the depletion region has two fermi level E_{Fn} and E_{Fp} that are flat across thetwo region such that they are seperately at equilibrium
2. Usually equilibrium between conduction band e- and valence band h+
3. Non-equilibrium, (shining light, injecting minority carriers in a PN junction), usually e- equilibrium in cond-b, h+ equilibrium in valence-b, quasi-equilibrium f^e and f^h, their own fermi-level.

Minority Carrier Density n_{P0}, and excess is what's there minus what's suppose to be there $n'_(x) = n_(x_P) - n_{P0}$. V is the bias voltage.

$$n'_(x_P) = n_{P0}(e^{V/V_T} - 1) \qquad n_{P0} = \frac{n_i^2}{N_a}$$
$$p'_(x_N) = p_{N0}(e^{V/V_T} - 1) \qquad p_{N0} = \frac{n_i^2}{N_d}$$

Excess carrier density as a function of x. Usually we put N-type at $x > 0$ and $x_N = x_P$ at origin.

$$n'(x) = n'(x_P)e^{(x-x_P)/L_n} p'(x) = \quad p'(x_N)e^{|x-x_N|/L_p}$$

Current for Excess Minority Carrier $\Delta\hat{q}$ ($\hat{q}$ is either n_p or p_n) Diffusion Equation, (some times they use Δ, δ, $\hat{q}'$). G_L is all other process. D is the diffusion coefficient.

$$\frac{d\Delta\hat{q}}{dt} = D_{(\mathrm{N/P})}\frac{d^2\Delta\hat{q}}{dx^2} - \frac{\Delta\hat{q}}{\tau_{(\mathrm{n/p})}} + G_L(x)$$

The steady state continuity equations for minority carriers are (L indicates diffusion length and the n'/p' indicates excess n/p carriers, D is the diffusion coefficient and τ is the mean collision time)

$$\frac{d^2p'}{dx^2} = \frac{p'}{L_p^2} \qquad \frac{d^2n'}{dx^2} = \frac{n'}{L_n^2}$$
$$L_p^2 = D_p\tau_p \qquad L_n^2 = D_n\tau_n$$

1.6.3 Current

Through Einstein relation $D_n/\mu_n = D_p/\mu_p = kT/q$, 0 current means constant fermi-level (note p-type E_F closer to valence, n to conduction)

$$J_n = \mu_n n \frac{dE_{F,n}}{dx} \qquad J_p = \mu_p p \frac{dE_{F,p}}{dx}$$

Total current $I_{tot} = I_p + I_n$

$$I = qA\left[\frac{D_p p_{n0}}{|W_{tn} - W_n|} + \frac{D_n n_{p0}}{|W_{tp} - W_p|}\right]\left[e^{\left(\frac{V}{V_T}\right)} - 1\right]$$

At $V = 0$, $I_{tot} = 0$, $I_{drift} = I = -I_0$, and so $I_{drift} = I_0$. Here n is the ideality factor and n_p describes the electron concentration on p side and vice versa.

$$I_0 = qA\left[\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right] \qquad I = I_0(e^{\frac{V}{nV_T}} - 1)$$

It just happens that

$$\begin{aligned} I_0 &= I_{h+,leak} + I_{e-,leak} \\ &= Aqn_i^2\left(\frac{D_P}{L_P N_D} + \frac{D_N}{L_N N_A}\right) \end{aligned}$$

1.6.4 Forward Bias

This is the Minority Carrier diffusion Current, Minority so no drift component.

$$\frac{J_{pN}}{q} = -D_p\frac{dp'(x)}{dx} = \frac{D_p p_{N0}}{L_p}(e^{V/V_T} - 1)e^{\frac{-(x-x_N)}{L_p}}$$
$$\frac{J_{nP}}{q} = -D_n\frac{dn'(x)}{dx} = \frac{D_n n_{P0}}{L_n}(e^{V/V_T} - 1)e^{\frac{-(x-x_P)}{L_n}}$$

1.6.5 Reverse Bias

Diffusion drops first, as h+ recombine with e- in n-region, equal e- injected into the region, e- drift current provide in the n-region to balance the holes into the region, Current same as forward bias. Before breakdown leakage current is just I_0

$$V_{break} = \frac{\varepsilon_s E_{crit}^2}{2qN}$$

1.6.6 Shocky Diode

Metal Contact

$$I_0 = AKT^2 e^{-V_{bi}/V_T}$$

1.6.7 Usages

Solar Cell Power

$$P = IV = \left[I_{sc} - I_0 exp\left(\frac{qV}{kT}\right)\right]V$$

1.7 Bipolar Junction Transistor

Practically just two PN Junctions doped such that its N+, P, N junction. Therefore still follows

$$\frac{d^2n'}{dx^2} = \frac{n'}{L_B^2}$$

Currents, $\beta = I_C/I_B$, $\alpha = I_C/I_E$

$$\beta = \frac{I_C}{I_B} = \frac{G_E}{G_B} = \frac{D_B W_E N_E n_{iB}^2}{D_E W_B N_B n_{iE}^2}$$

The emitter efficiency is defined as

$$\gamma_E = \frac{I_E - I_B}{I_E} = \frac{I_C}{I_C + I_B} = \frac{1}{1 + G_B/G_E}$$

$$\begin{aligned} I_C &= A_E q D_B \frac{dn}{dx} = A_E q D_B \frac{n'(0)}{W_B} \\ &= A_E q \frac{D_B n_{iB}^2}{W_B N_B}(e^{(V_{BE}/V_T)} - 1) \\ &= I_S(e^{(V_{BE}/V_T)} - 1) \end{aligned}$$

In this special case we can say (G_B is base dopant atoms per unit area / D_B or $\frac{1}{D_B}\int_0^{W_B} N_B dx$

$$G_B = \frac{n_i^2 N_B}{n_{iB}^2 D_B} W_B = \frac{n_i^2 p}{n_{iB}^2 D_B} W_B$$

And Base Current

$$\begin{aligned} I_B &= A_E q \frac{D_E n_{iE}^2}{W_E N_E}(e^{(V_{BE}/V_T)} - 1) \\ I_B &= \frac{I_S}{\beta_F}(e^{(V_{BE}/V_T)} - 1) \end{aligned}$$

1.8 MOSCAP

Surface Voltage ϕ_s is the amount the vacuum level of metal and silicon needs to bend to align their fermi-level (their work function, note that the metal $E_f = E_c$). The edge of depletion($<$) and inversion($>$) is $\phi_s = 2\phi_b$. Note that this equality is also when the max depletion width occurs. $\varepsilon_s = 11.8 \times \varepsilon_0$ and $\varepsilon_0 = 8.85 \times 10^{-14}$

$$\begin{aligned} \phi_b &= E_i - E_f & E_i &= \frac{1}{2}(E_c - E_v) \\ &= kT \ln\frac{N_A}{n_i} & W_{dep} &= \sqrt{\frac{2\varepsilon_s\phi_s}{qN_a}} \end{aligned}$$

Field E_s

$$E_s = \sqrt{\frac{qN_a|\phi_s|}{\varepsilon_s}}$$

Flat-Band voltage is required to bring the two vacuum level together such that surface voltage = 0. For metal $E_f = E_c$,

$$V_{fb} = \psi_g - \psi_s,$$

The difference between fermi level V_g may also be found.

$$V_g = V_{fb} + \phi_s + V_{ox} \qquad V_{ox} = \frac{Q_{sub}}{C_{ox}}$$

The bias condition is accumulation when voltage less than V_{fb} is applied

$$Q_{acc} = -C_{ox}(V_g - V_{fb})$$

We can find the threshold voltage by combining the previous information, $C_{ox} = \varepsilon_{ox}/t_{ox}$ (Note nmos/pmos have p/n substrate)

$$V_t = V_g(\phi_s = 2\phi_B) = V_{fb} + \phi_s + \frac{\sqrt{qN_a 2\varepsilon_s \phi_s}}{C_{ox}}$$

Capacitance per unit area, $\varepsilon_{ox} = 3.9 \times \varepsilon_0$. Normalized Cap is C/C_{ox}. At high frequencies, charges cannot keep up in the depletion region, so the capacitance never rises suddenly after dropping.

$$C = C_s || C_{ox} \qquad C_{sth} = \varepsilon_{sth}/t_{sth}$$

1.9 FET

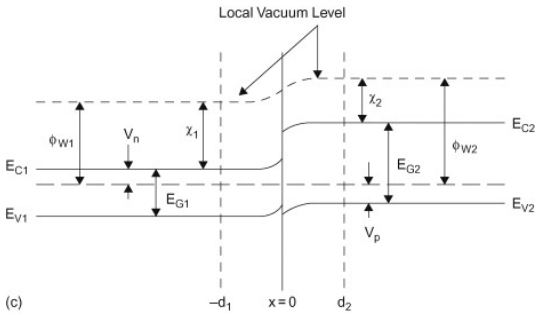
Pinch off voltage (Large V_D such that depletion increases, inversion layer reduced to zero)

$$V_{pinch} = \frac{qN_D h^2}{2\varepsilon}$$

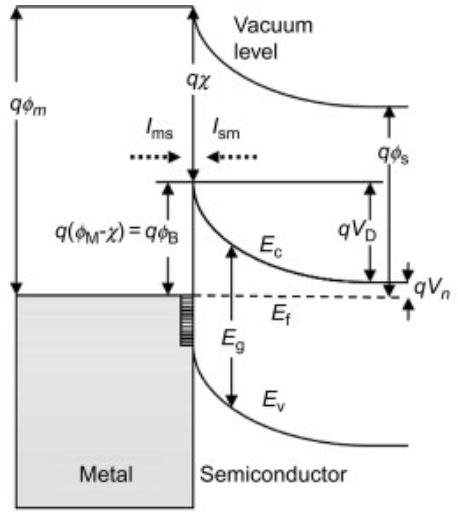
Gate Bias and Channel width

$$W = \sqrt{\frac{2\varepsilon(V_{bi} - V_{gs})}{qN_A}}$$

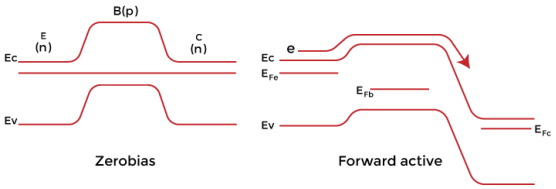
Appendix: Diagrams



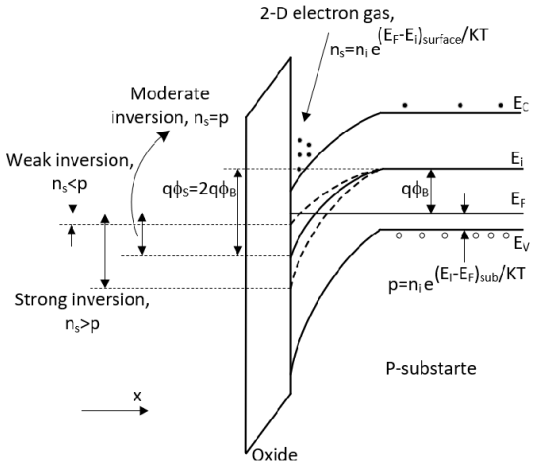
PN-Junction



Metal and N-Type Schottky



NPN BJT, E-B-C



MOSCAP